Michael Ajao-Olarinoye, PhD, AFHEA

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Personal Profile

AI scientist and engineer specialising in applied machine learning, generative AI and scientific modelling. Research Fellow at Coventry University and Lecturer in Computing, with a PhD in Computational Science and Mathematical Modelling. Experienced in developing and deploying AI systems for healthcare, student engagement and business automation, alongside peer-reviewed research and postgraduate supervision. Experience also includes AI solutions and technical training delivered through miolajtech.

Research Expertise & Technical Skills

- **Machine Learning & AI:** Generative AI, language model fine-tuning, AI agents, deep learning, natural language processing, computer vision; PyTorch, TensorFlow, scikit-learn, LightGBM, XGBoost, CatBoost
- **Scientific Modelling:** Time-series forecasting, physics-informed neural networks, hybrid and graph-based models, statistical modelling, survival analysis, optimisation
- **Programming & Data:** Python, R, Julia, SQL; Pandas, NumPy, data pipelines and visualisation
- **Software & Cloud:** FastAPI, Flask, Docker, Git; AWS, Google Cloud, Microsoft Azure; JavaScript, React, Flutter
- **Research & Teaching:** Research design, academic writing, peer review, student supervision, curriculum development, project management; LaTeX

Education

- **Coventry University** *May 2021 – March 2025*
  PhD in Computational Science and Mathematical Modelling Coventry, United Kingdom
  - Thesis: Data-Driven Algorithms for Forecasting Critical Care Demand during the COVID-19 Pandemic.
  - Developed deep learning and physics-informed models to forecast COVID-19 hospital admissions and intensive care demand in England.
  - Established a cardiovascular modelling collaboration with the National University of Sciences and Technology, Pakistan, and mentored postgraduate students.
- **Coventry University** *January 2020 – January 2021*
  MSc in Data Science and Computational Intelligence Coventry, United Kingdom
  - Dissertation (Distinction): Plant Disease Detection: A Cross-Platform Mobile Application Using Artificial Neural Networks.
- **Kwara State University (KWASU)** *September 2011 – July 2016*
  BEng in Electrical and Electronics Engineering Malete, Nigeria
  - Second Class Honours. Final-year project: Designed and built a low-power FM transmitter with a 2 km range.

Work Experience

- **Coventry University, Strategic Planning and Insight Office** *September 2026 – Present*
  Research Fellow, EVOLVE AI Project Coventry, United Kingdom
  - Lead AI research for EVOLVE AI, developing models to predict student engagement, academic risk and course withdrawal to support early intervention.
  - Built an analytics pipeline combining attendance, online learning activity and assessment records across undergraduate and postgraduate courses.
  - Compared machine learning and deep learning models for student risk prediction, evaluating performance across courses and prediction periods.
  - Tuned models to prioritise identifying at-risk students, developed explanations for predictions, and communicated findings through stakeholder reports and executive briefings.
- **Coventry University, CSMM** *January 2026 – August 2026*
  Research Fellow, Centre for Computational Science & Mathematical Modelling Coventry, United Kingdom
  - Led applied machine learning and generative AI research for the Innovate UK-funded HOPE-MOVE project, supporting digital health and self-management programmes.
  - Developed and deployed models to predict participant dropout, integrating early and weekly risk predictions into the Hope Dashboard to support participant retention.
  - Analysed participant behaviour and discussion content to identify early engagement patterns associated with programme completion.
  - Fine-tuned language models to generate empathetic facilitator responses guided by motivational interviewing principles.
  - Implemented crisis detection, mental health signposting and local model deployment to support safe, privacy-conscious AI use and human review.

- **QAHE Ltd** *February 2025 – Present*
  Lecturer in Computing Birmingham, United Kingdom
  - Teach programming, data science and machine learning through QAHE's partnerships with Ulster University and Southampton Solent University.
  - Supervised more than 10 MSc dissertations at Ulster University, supporting research design, methodology and thesis completion.
  - Develop teaching materials and assessments, provide inclusive academic support, and contribute to curriculum development and quality assurance.
- **The Futureworks (Coventry University Group)** *April 2021 – January 2025*
  Graduate Teaching Assistant (Part-time) Coventry, United Kingdom
  - Developed and delivered tutorials and workshops in information retrieval, big data analytics, deep learning and computer vision.
  - Assessed student work, provided constructive feedback and offered individual academic support.
- **Tammy Tech** *March 2020 – April 2021*
  Software Developer Lagos, Nigeria
  - Developed web and mobile applications using JavaScript and Flutter, expanding client product offerings by 30%.
  - Worked with clients to translate requirements into software solutions, using agile methods to deliver projects on time.
  - Integrated APIs and databases to support web and mobile application functionality.
- **Verificar Compliance** *September 2018 – January 2020*
  Business Systems Analyst Lagos, Nigeria
  - Analysed business processes and developed IT solutions that improved operational efficiency by 15%.
  - Built performance dashboards and reporting systems, coordinating with technical teams and stakeholders to align solutions with business needs.

Publications

- Ur Rehman, M. A., Ekici, Ö., Erdener, Ş. E., **Ajao-Olarinoye, M.**, Kuchumov, A. G., & Jia, F. (2026). Wall shear stress in coronary arteries and grafts using computational fluid dynamics and physics-informed neural networks. *Physics of Fluids*, 38(9), 091905. doi:10.1063/5.0335500
- Ur Rehman, M. A., Ekici, Ö., Farooq, M. A., Butt, K., **Ajao-Olarinoye, M.**, Wang, Z., & Liu, H. (2025). Fluid–structure interaction analysis of pulsatile flow in arterial aneurysms with physics-informed neural networks and computational fluid dynamics. *Physics of Fluids*, 37(3), 031913. doi:10.1063/5.0259296
- Fatima, N., Hamza, M., Farooq, M. A., & **Ajao-Olarinoye, M.** (2026). Modeling of double descending thoracic aortic aneurysms using computational fluid dynamics (CFD) and residual-based physics-informed neural networks (ResNets-PINNs). *Pramana – Journal of Physics*. Accepted, in press.
- **Ajao-Olarinoye, M.**, Babatunde, A., Sadeghimanesh, A., He, F., & England, M. (2026). Physics-informed multi-quantile forecasting for risk-averse NHS critical-care surge allocation. *Proceedings of the 25th UK Workshop on Computational Intelligence (UKCI 2026)*, Coventry. Springer. Accepted, in press; presented 10 September 2026.
- **Ajao-Olarinoye, M.**, Palade, V., He, F., Wark, P. A., Mukandavire, Z., & Mousavi, S. (2025). A hybrid physics-informed neural network–SEIRD model for forecasting COVID-19 intensive care unit demand in England. In U. Onyekpe, V. Palade, & M. A. Wani (Eds.), *Recent Advances in Deep Learning Applications* (pp. 231–256). Chapman & Hall/CRC (Taylor & Francis). doi:10.1201/9781003570882-15
- **Ajao-Olarinoye, M.**, Palade, V., Mousavi, S., He, F., & Wark, P. A. (2023). Deep learning based forecasting of COVID-19 hospitalisation in England: A comparative analysis. In *2023 International Conference on Machine Learning and Applications (ICMLA)* (pp. 1344–1349). IEEE. doi:10.1109/ICMLA58977.2023.00203

Manuscripts & Preprints

- **Ajao-Olarinoye, M.**, Palade, V., He, F., Wark, P. A., Mousavi, S., & Mukandavire, Z. (2025). MSAGAT-Net: Multi-scale temporal adaptive graph attention for efficient spatiotemporal epidemic forecasting. Preprint, under review. SSRN. doi:10.2139/ssrn.5719497
- Ur Rehman, M. A., Temimi, H., **Ajao-Olarinoye, M.**, Laadhari, A., Riahi, M. K., & Kissami, I. (Under review). Multiphysics fluid–structure interaction modeling and physics-informed neural network surrogate for morphology-dependent stress localization in thoracoabdominal aneurysms.
- Zahra, **Ajao-Olarinoye, M.**, Farooq, M. A., & Khan, F. (Submitted). Comparative analysis of numerical schemes and physics-informed neural networks for two-dimensional dam-break problems with source terms under different initial dam breaking conditions.
- Jonah, S., Moses, Q., Babatunde, A., **Ajao-Olarinoye, M.**, & Bammeke, D. (2026). LLMs for agentic home energy management. Preprint. arXiv:2607.04569
- Jonah, S., Akinseloyin, O., **Ajao-Olarinoye, M.**, & Babatunde, A. (2026). Controlled evaluation of graph and multimodal augmentation in RAG for document question answering. Preprint. arXiv:2607.16604

Talks and Posters

- **Invited talk**, Centre for Doctoral Training in Artificial Intelligence (CDT in AI) Launch Event, Coventry University, 20 October 2026. Talk title: *From Student to Success: An AI Alumni Journey*.
- **Oral presentation**, 25th UK Workshop on Computational Intelligence (UKCI 2026), Coventry, 10 September 2026. Talk title: *Physics-Informed Multi-Quantile Forecasting for Risk-Averse NHS Critical-Care Surge Allocation*.
- **Invited speaker**, 3rd edition of "Future Algorithms", online, 29–30 April 2026. Talk title: *Algorithmic Training Strategies for Physics-Informed Neural Network Wall Shear Stress Prediction in Cardiovascular Hemodynamics*.
- *Physics-Informed Neural Networks for Fluid-Structure Interaction Analysis of Arterial Aneurysms*, poster presented at the 2nd International Conference of "Future Algorithms", online, 11–12 June 2025 (**First Prize**, poster competition).
- **Talk**, 2nd CSMM Annual Conference, Coventry University, May 2024: *Physics-Informed Neural Networks for Modelling Infectious Disease Dynamics: A Case Study of COVID-19 in England*.
- **Poster**, IEEE ICMLA, 2023: *Deep Learning Based Forecasting of COVID-19 Hospitalisation in England: A Comparative Analysis*.

Leadership and Volunteering

- **African Institute for Artificial Intelligence (AI4AI)** — *January 2024 – Present*
  Software Engineering Team Lead, AI Literacy team (Volunteer) — Remote
- **CodeYourFuture, Birmingham, United Kingdom** — *December 2020 – Present*
  Lecturer — Remote
- **National Youth Service Corps (NYSC), Nigeria** — *November 2016 – October 2017*
  Project Manager for Gender Vanguard CDS — On-site
- **Nigeria Union of Engineering Students Association (NUESA), Kwara State University** — *June 2014 – June 2015*
  Vice President — On-site

Certifications

- **Chartered Management Institute (CMI)** - 2021: CMI Level 7 Certificate in Strategic Management and Leadership; CMI Level 7 Award in Professional Consulting

Honours and Awards

- **UK Global Talent**
  Endorsed by the Royal Academy of Engineering (Exceptional Promise) — September 2026
- **2nd International Conference of "Future Algorithms"**
  First Prize, Poster Competition — July 2025
- **CSMM Conference**
  Best Reviewer Award — June 2024
- **Coventry University**
  PhD Studentship Award (fully funded) — May 2021
- **Kwara State University (KWASU), Entrepreneurship and Innovation Challenge**
  Third Place in Community Development Project — October 2017

Professional Memberships & Academic Service

- Programme Committee Member, IEEE International Conference on Machine Learning and Applications (ICMLA), 2024
- Associate Fellow of the Higher Education Academy (AFHEA) - 2024 – Present
- IEEE Student Member - 2021 – Present
- Royal Statistical Society e-Student Member - 2024 – Present

Languages

English (professional proficiency), French (elementary proficiency), Yoruba (native proficiency).